

SECTION A [20*0.5 = 10 Marks]

Write your answers in the table below

Q #	1	2	3	4	5	6	7	8	9	10
Answer										
Q #	11	12	13	14	15	16	17	18	19	20
Answer										

1. Which of the following is correct regarding asymmetric cryptography?

- A) Encryption and decryption take the same amount of time.
- B) The same key is used for encryption and decryption.
- C) Cryptographic operations are one-way, and not reversible.
- D) Different keys are used for encryption and decryption.

2. Which of these may be used for encryption/decryption, Digital signature, and Key exchange?

- A) RSA and DES
- B) RSA and Diffie-Hellman
- C) DES and Diffie-Hellman
- D) RSA and Elliptic Curve

3. In RSA, the Euler Totient Function, $\phi(n)$, is

- A) private and calculated
- B) public and chosen
- C) private and chosen
- D) public and calculated

4. Bubu digitally signs a message and sends it to Dudu. Verification of the signature by Dudu requires

- A) Bubu's Public Key
- B) Dudu's Public Key
- C) Dudu's Private Key
- D) Bubu's Private Key

5. Using public key cryptography, X adds a digital signature σ to message M, encrypts , and sends it to Y, where it is decrypted. Which one of the following sequences of keys is used for the operations?

- A) Encryption: X's private key followed by Y's private key; Decryption: Y's public key followed by X's public key
- B) Encryption: X's private key followed by Y's public key; Decryption: Y's public key followed by X's private key
- C) Encryption: X's public key followed by Y's private key; Decryption: Y's public key followed by X's private key
- D) Encryption: X's private key followed by Y's public key; Decryption: Y's private key followed by X's public key

6. In the RSA public key cryptosystem, the private and public keys are (e,n) and (d,n) respectively,

where $n=p*q$ and p and q are large primes. Besides, n is public and p and q are private. Let M be an integer such that $0 < M < \Phi(n)$ and $\Phi(n) = (p-1)(q-1)$. Now consider the following equations.

- I. $M' = M^e \bmod n$ $M = (M')^d \bmod n$
- II. $ed \equiv 1 \bmod n$
- III. $ed \equiv 1 \bmod \Phi(n)$
- IV. $M' = M^e \bmod \Phi(n)$ $M = (M')^d \bmod \Phi(n)$

Which of the above equations correctly represent RSA cryptosystem?

- A) (I) and (II)
- B) (I) and (III)
- C) (II) and (IV)
- D) (III) and (IV)

7. Which of the following is not a property of public key cryptography?

- A) Uses different keys
- B) Every user has a unique private key
- C) Public key can be same for number of users
- D) Cannot deduce private key from public key

8. Which of the following is correct about RSA scheme in public key cryptography?

- A) It is efficient for decryption
- B) Lengthy keys are required for more security
- C) It is not efficient for encryption
- D) It is not popular

9. Which of the following is an asymmetric key cryptography algorithm?

- A) DES
- B) AES
- C) RC4
- D) ELGAMAL CRYPTOSYSTEM

10. A sender encrypts a message using recipient's public key in public key cryptography?

- A) True
- B) False

11. In public key cryptosystem which is kept as public?

- A) Encryption keys
- B) Decryption keys
- C) Encryption & Decryption keys
- D) None of the mentioned

12. Examples of hash functions are

- A) MD5
- B) SHA-1
- C) MD5 & SHA-1
- D) None of the mentioned

13. A cryptographic hash function has variable output length.

- A) True
- B) False

14. A Digital Signature is required for

- A) non repudiation of communication
- B) all email sending

C) all DHCP Server

D) FTP transactions

15. The security of the ElGamal cryptosystem is primarily based on the difficulty of which mathematical problem?

A) Integer factorization problem

B) Elliptic curve problem

C) Discrete logarithm problem

D) Prime number generation

16. What is a potential vulnerability if the same ephemeral key 'k' is reused in multiple ElGamal encryptions?

A) increases encryption speed

B) has no effect on security

C) lead to the compromise of the private key

D) simplifies key management

17. In the Diffie-Hellman Key Exchange, which of the following values must be agreed upon and shared publicly by both parties before exchanging keys?

A) Private keys

B) Shared secret key

C) Prime number p and generator g

D) Ciphertext

18. What role does a hash function play in the digital signature process?

A) It encrypts the message directly

B) It generates a short digest to be signed, ensuring integrity

C) It verifies the identity of the recipient

D) It creates a key for symmetric encryption

19. What is the Discrete Logarithm Problem (DLP) in modular arithmetic?

A) Finding the logarithm of a number using base 10 in a finite field

B) Finding an integer x such that $g^x \equiv h \pmod{p}$ given g , h , and prime p .

C) Solving $x^2 + y^2 = z^2$ over integers

D) Calculating the inverse of a number modulo p

20. In ECC, what is the term for the point at which a straight line intersects an elliptic curve at three points that sum to zero?

A) Base Point

B) Inverse Point

C) Point at Infinity

D) Neutral Point

SECTION B []

Note :

1. Write answers in the table below.
2. For each question, do not write answers directly. Use appropriate methods and algorithms such as the Extended Euclidean Algorithm (EEA) and the Square and multiply method wherever applicable.
3. If you need additional space, use the extra sheets provided at the end of the question paper.

Question	Answer
1.	(i)
	(ii)
	(iii)
2.	
3.	
4.	

STUDENT ID:

- 1.** Tom and Jerry wish to use RSA to communicate securely. Tom chooses public key (e, n) as $(7, 247)$ and Jerry chooses public key (e, n) as $(5, 221)$

- (i) Calculate Tom's Private Key.

[2Marks]

Space for Calculation:

(ii) Calculate Jerry's Private Key.

[2Marks]

Space for Calculation:

STUDENT ID:

- (iii) What will be the cipher text sent by Tom to Jerry, if Tom wishes to send Message 5 to Jerry. **[1 Marks]**

Space for Calculation:

2. Bob sends an ElGamal-encrypted message to Alice using the same secret key k_e for all messages Alice receives two ciphertexts:

- $C1=(Y1=6, Y2=17)$ corresponding to the plaintext $x1=21$.
- $C2=(Y1=6, Y2=25)$ corresponding to an unknown plaintext $x2$.

The ElGamal encryption parameters are: Prime modulus $p=31$, Generator (Primitive root) $\alpha=3$, Public key $\beta=18$. Given this information, determine the second plaintext $x2$.
[3 Marks]

Space for Calculation:

STUDENT ID:

3. Consider the elliptic curve $E: y^2 \equiv x^3 + 2x + 2 \pmod{17}$, and let $P = (5, 1)$ be a point on the curve. Compute the point $2P$ (i.e., the result of doubling the point P) on the given elliptic curve. **[2 Marks]**

Space for Calculation:

